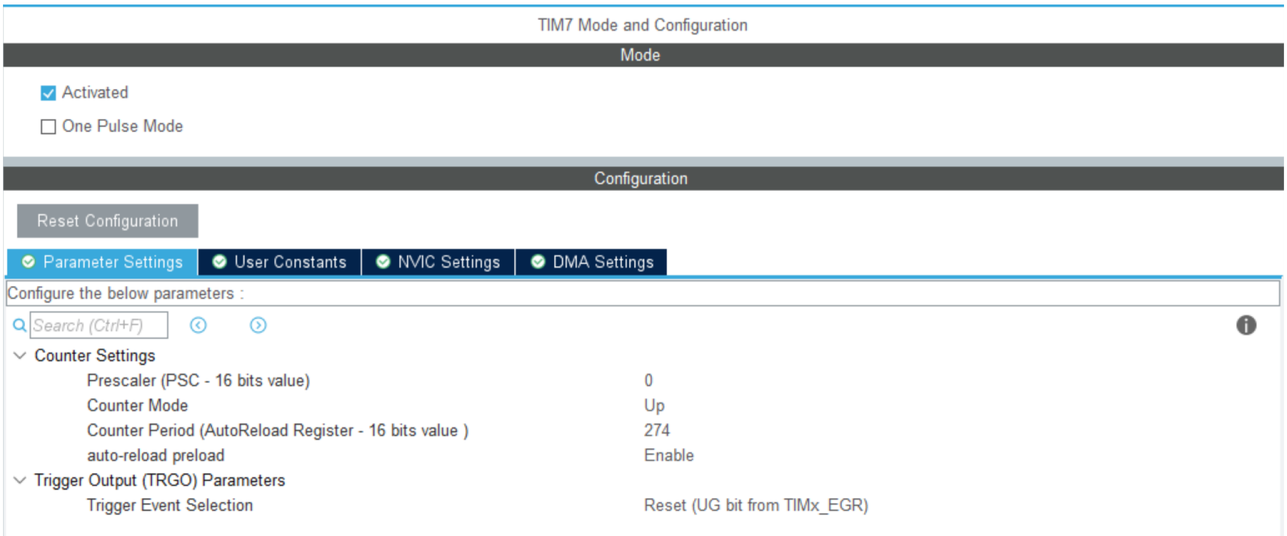
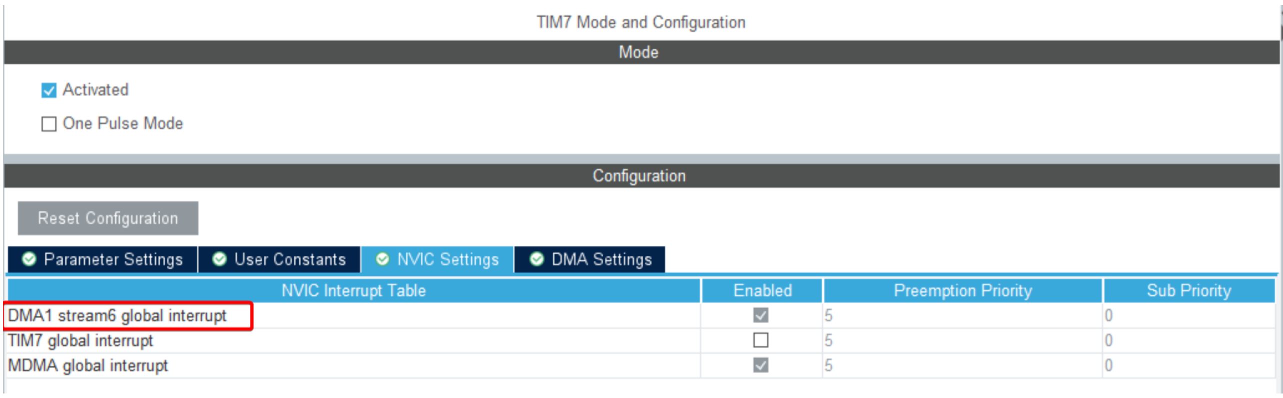
1. 文档说明

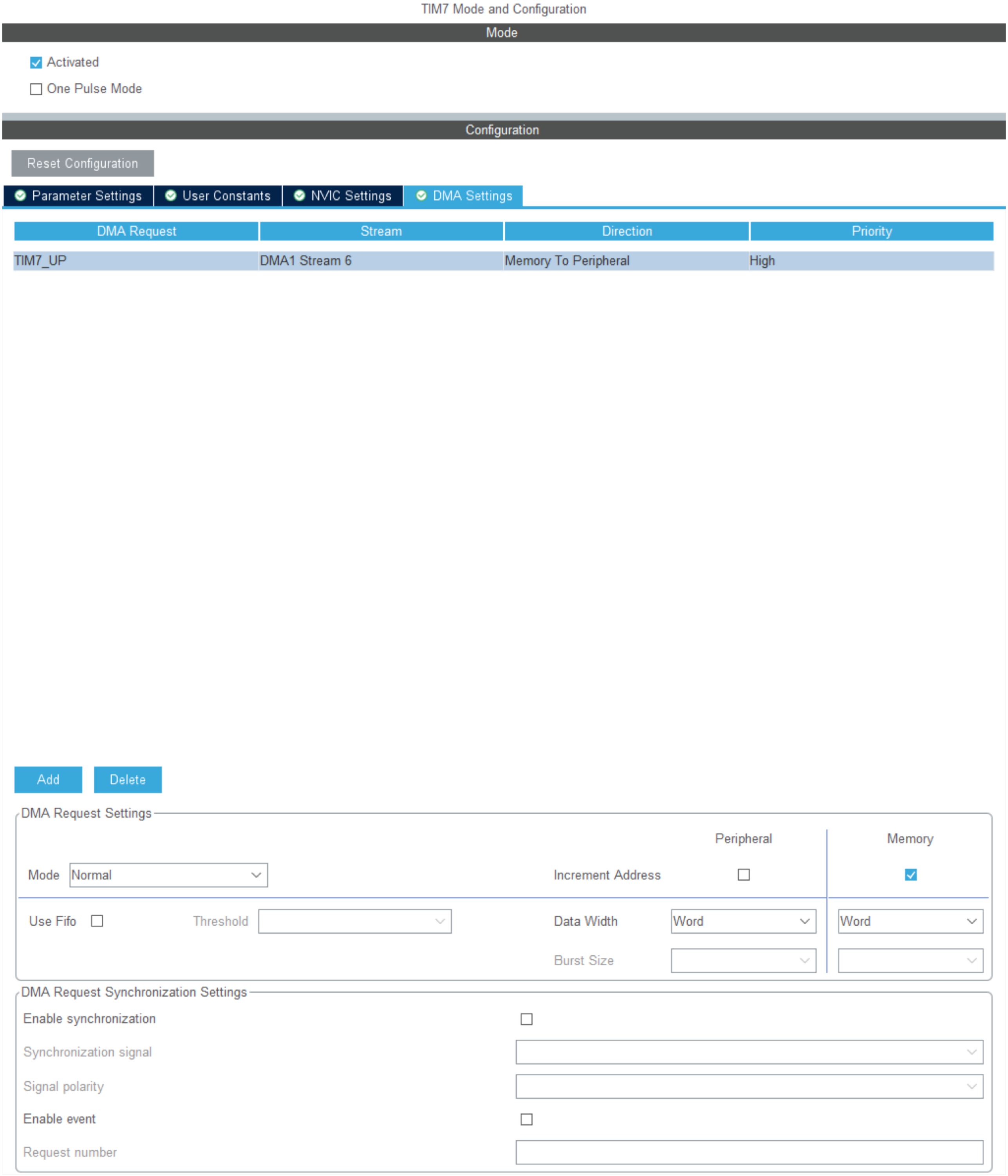
- 本demo针对新增片上外设，需要设备驱动情况时，工程修改参考
- 本demo拟实现如下功能：
  - 使用TIM7，定时周期为1us，每次定时溢出触发一次DMA数据搬运（memory->peripheral）
  - TIM+DMA组合方式，可以控制GPIO以任意频率/波特率输出数据(本demo定时周期为1us，故波特率为1Mbps)，从而实现使用GPIO模拟UART、I2C、SPI等串行总线通信
  - 基础分支hash: 1b43d79b5e6d06a3eaa587a3097d763b7057e294
  - demo分支hash: 9d09b3435ad3ef6e45b67527b28c197344b30750

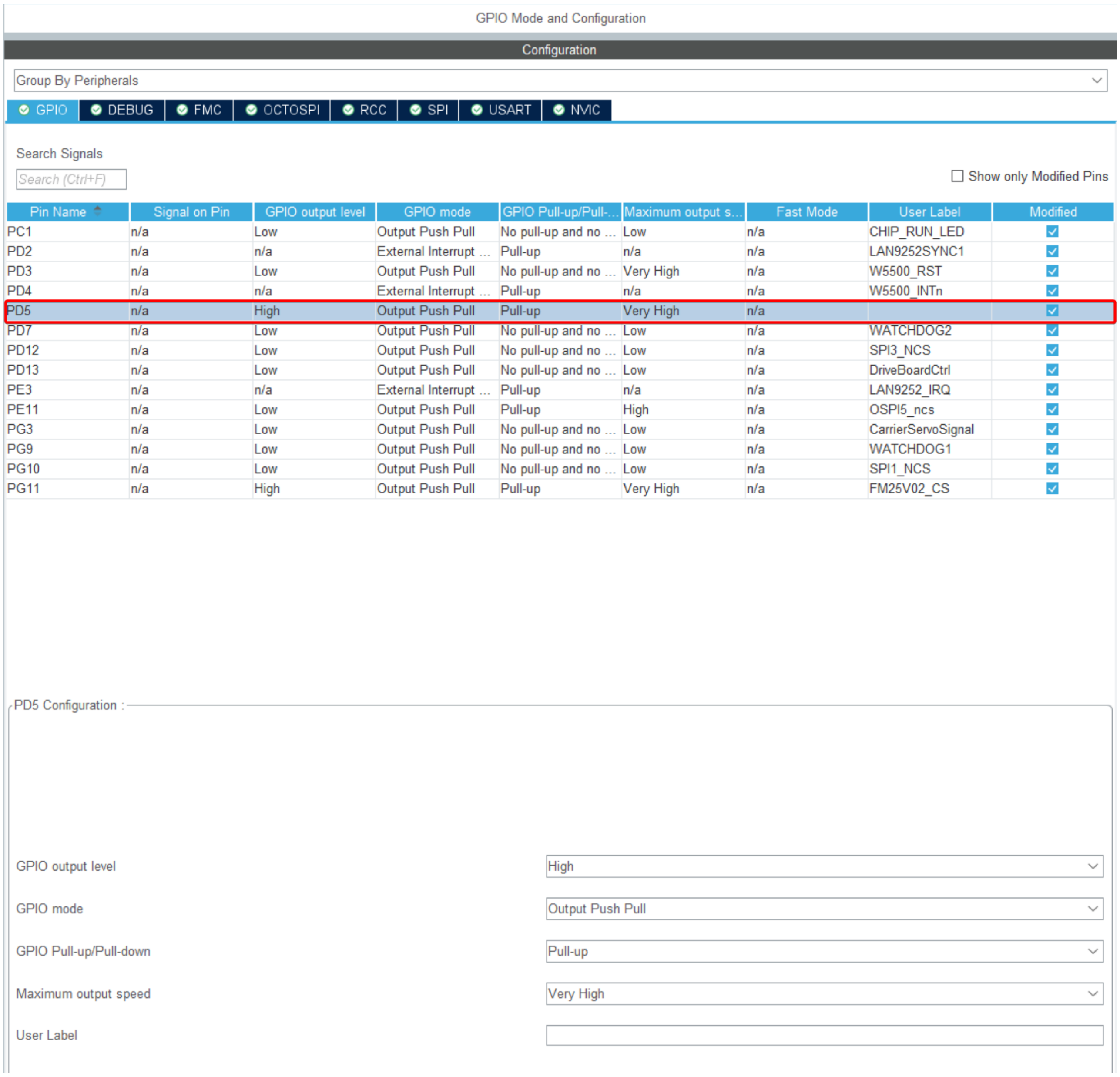
2. 修改方式

- 使用STM32CubeMX工具配置TIM7、DMA、GPIOD_5



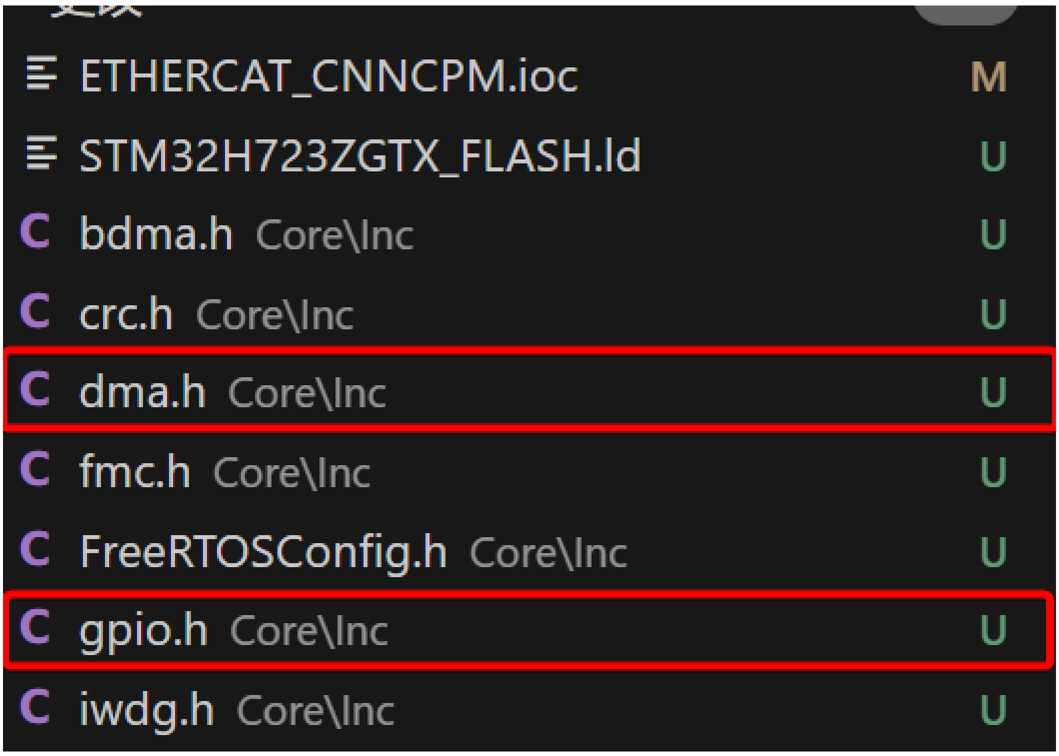


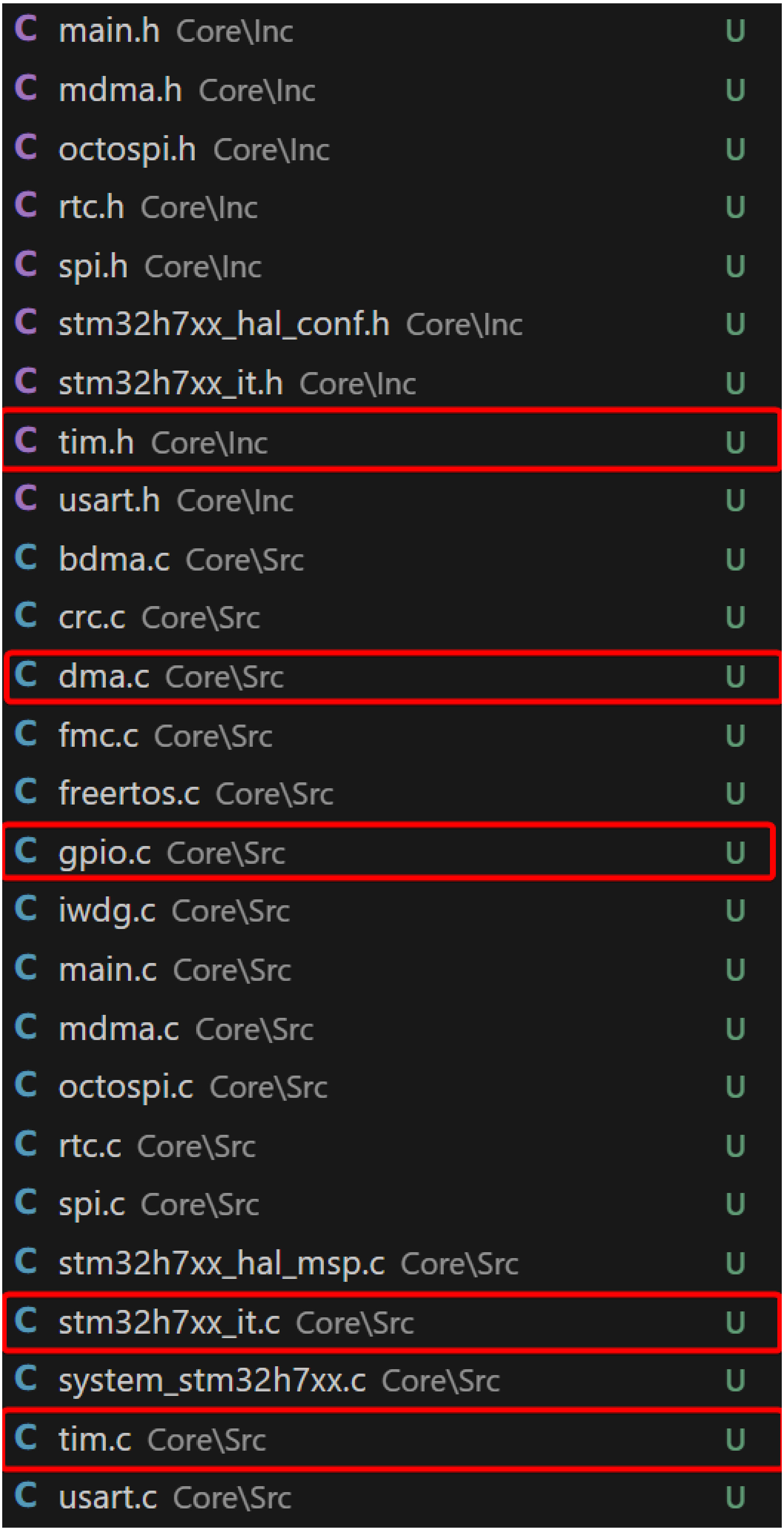

TIM7 Mode and Configuration
Mode
Activated
One Pulse Mode
Configuration
Reset Configuration
Parameter Settings
User Constants
NVIC Settings
DMA Settings
DMA Request
Stream
Direction
Priority
TIM7_UP
DMA1 Stream 6
Memory To Peripheral
High
Add
Delete
DMA Request Settings
Peripheral
Memory
Mode
Normal
Increment Address
Use Fifo
Threshold
Data Width
Word
Word
Burst Size
DMA Request Synchronization Settings
Enable synchronization
Synchronization signal
Signal polarity
Enable event
Request number



Pin Name	Signal on Pin	GPIO output level	GPIO mode	GPIO Pull-up/Pull-...	Maximum output s...	Fast Mode	User Label	Modified
PC1	n/a	Low	Output Push Pull	No pull-up and no ...	Low	n/a	CHIP_RUN_LED	☑
PD2	n/a	n/a	External Interrupt ...	Pull-up	n/a	n/a	LAN9252SYNC1	☑
PD3	n/a	Low	Output Push Pull	No pull-up and no ...	Very High	n/a	W5500_RST	☑
PD4	n/a	n/a	External Interrupt ...	Pull-up	n/a	n/a	W5500_INTn	☑
PD5	n/a	High	Output Push Pull	Pull-up	Very High	n/a		☑
PD7	n/a	Low	Output Push Pull	No pull-up and no ...	Low	n/a	WATCHDOG2	☑
PD12	n/a	Low	Output Push Pull	No pull-up and no ...	Low	n/a	SPI3_NCS	☑
PD13	n/a	Low	Output Push Pull	No pull-up and no ...	Low	n/a	DriveBoardCtrl	☑
PE3	n/a	n/a	External Interrupt ...	Pull-up	n/a	n/a	LAN9252_IRQ	☑
PE11	n/a	Low	Output Push Pull	Pull-up	High	n/a	OSPI5_ncs	☑
PG3	n/a	Low	Output Push Pull	No pull-up and no ...	Low	n/a	CarrierServoSignal	☑
PG9	n/a	Low	Output Push Pull	No pull-up and no ...	Low	n/a	WATCHDOG1	☑
PG10	n/a	Low	Output Push Pull	No pull-up and no ...	Low	n/a	SPI1_NCS	☑
PG11	n/a	High	Output Push Pull	Pull-up	Very High	n/a	FM25V02_CS	☑

- 生成代码，找到与TIM7、DMA相关的头文件和源文件，对比修改工程前的相应文件，并拷贝差异部分




C main.h Core\Inc U
C mdma.h Core\Inc U
C octospi.h Core\Inc U
C rtc.h Core\Inc U
C spi.h Core\Inc U
C stm32h7xx_hal_conf.h Core\Inc U
C stm32h7xx_it.h Core\Inc U
C tim.h Core\Inc U
C usart.h Core\Inc U
C bdma.c Core\Src U
C crc.c Core\Src U
C dma.c Core\Src U
C fmc.c Core\Src U
C freertos.c Core\Src U
C gpio.c Core\Src U
C iwdg.c Core\Src U
C main.c Core\Src U
C mdma.c Core\Src U
C octospi.c Core\Src U
C rtc.c Core\Src U
C spi.c Core\Src U
C stm32h7xx_hal_msp.c Core\Src U
C stm32h7xx_it.c Core\Src U
C system_stm32h7xx.c Core\Src U
C tim.c Core\Src U
C usart.c Core\Src U

- 拷贝差异部分至相应的原代码文件，并根据业务需求确定句柄、函数是否添加到头文件中

> dma.c
> ```c
> void MX_DMA_Init(void)
> {
>
>   /* DMA controller clock enable */
>   __HAL_RCC_DMA1_CLK_ENABLE();
>   __HAL_RCC_DMA2_CLK_ENABLE();
>
>   /* DMA interrupt init */
>   /* DMA1_Stream0_IRQn interrupt configuration */
>   HAL_NVIC_SetPriority(DMA1_Stream0_IRQn, 5, 0);
>   HAL_NVIC_EnableIRQ(DMA1_Stream0_IRQn);
>   /* DMA1_Stream1_IRQn interrupt configuration */
>   HAL_NVIC_SetPriority(DMA1_Stream1_IRQn, 5, 0);
>   HAL_NVIC_EnableIRQ(DMA1_Stream1_IRQn);
>   /* DMA1_Stream2_IRQn interrupt configuration */
>   HAL_NVIC_SetPriority(DMA1_Stream2_IRQn, 5, 0);
>   HAL_NVIC_EnableIRQ(DMA1_Stream2_IRQn);
>   /* DMA1_Stream3_IRQn interrupt configuration */
>   HAL_NVIC_SetPriority(DMA1_Stream3_IRQn, 5, 0);
>   HAL_NVIC_EnableIRQ(DMA1_Stream3_IRQn);
>   /* DMA1_Stream4_IRQn interrupt configuration */
>   HAL_NVIC_SetPriority(DMA1_Stream4_IRQn, 5, 0);
>   HAL_NVIC_EnableIRQ(DMA1_Stream4_IRQn);
>   /* DMA1_Stream5_IRQn interrupt configuration */
>   HAL_NVIC_SetPriority(DMA1_Stream5_IRQn, 5, 0);
>   HAL_NVIC_EnableIRQ(DMA1_Stream5_IRQn);
>   /* DMA1_Stream6_IRQn interrupt configuration */
>   HAL_NVIC_SetPriority(DMA1_Stream6_IRQn, 5, 0);
>   HAL_NVIC_EnableIRQ(DMA1_Stream6_IRQn);
>   /* DMA1_Stream7_IRQn interrupt configuration */
>   HAL_NVIC_SetPriority(DMA1_Stream7_IRQn, 5, 0);
>   HAL_NVIC_EnableIRQ(DMA1_Stream7_IRQn);
>   /* DMA2_Stream0_IRQn interrupt configuration */
>   HAL_NVIC_SetPriority(DMA2_Stream0_IRQn, 5, 0);
>   HAL_NVIC_EnableIRQ(DMA2_Stream0_IRQn);
>
> }
> ```

gpio.c

```c
  HAL_GPIO_Init(GPIOD, &GPIO_InitStruct);

  /*Configure GPIO pin : PtPin */
  GPIO_InitStruct.Pin = W5500_RST_Pin;
  GPIO_InitStruct.Mode = GPIO_MODE_OUTPUT_PP;
  GPIO_InitStruct.Pull = GPIO_NOPULL;
  GPIO_InitStruct.Speed = GPIO_SPEED_FREQ_VERY_HIGH;
  HAL_GPIO_Init(W5500_RST_GPIO_Port, &GPIO_InitStruct);

  /*Configure GPIO pin : PD5 */
  GPIO_InitStruct.Pin = GPIO_PIN_5;
  GPIO_InitStruct.Mode = GPIO_MODE_OUTPUT_PP;
  GPIO_InitStruct.Pull = GPIO_PULLUP;
  GPIO_InitStruct.Speed = GPIO_SPEED_FREQ_VERY_HIGH;
  HAL_GPIO_Init(GPIOD, &GPIO_InitStruct);

  /*Configure GPIO pin : PtPin */
  GPIO_InitStruct.Pin = FM25V02_CS_Pin;
  GPIO_InitStruct.Mode = GPIO_MODE_OUTPUT_PP;
  GPIO_InitStruct.Pull = GPIO_PULLUP;
  GPIO_InitStruct.Speed = GPIO_SPEED_FREQ_VERY_HIGH;
  HAL_GPIO_Init(FM25V02_CS_GPIO_Port, &GPIO_InitStruct);

  /* EXTI interrupt init*/
  HAL_NVIC_SetPriority(EXTI2_IRQn, 7, 0);
  HAL_NVIC_EnableIRQ(EXTI2_IRQn);

  HAL_NVIC_SetPriority(EXTI3_IRQn, 5, 0);
  HAL_NVIC_EnableIRQ(EXTI3_IRQn);

  HAL_NVIC_SetPriority(EXTI4_IRQn, 5, 0);
  HAL_NVIC_EnableIRQ(EXTI4_IRQn);
```

tim.c

```c
#include "tim.h"

/* USER CODE BEGIN 0 */

/* USER CODE END 0 */

TIM_HandleTypeDef htim1;
TIM_HandleTypeDef htim2;
TIM_HandleTypeDef htim6;
TIM_HandleTypeDef htim7;        You, 5分钟前 • Uncommitted changes
DMA_HandleTypeDef hdma_tim7_up;
```

```c
/* TIM7 init function */
void MX_TIM7_Init(void)
{

  /* USER CODE BEGIN TIM7_Init 0 */

  /* USER CODE END TIM7_Init 0 */

  TIM_MasterConfigTypeDef sMasterConfig = {0};

  /* USER CODE BEGIN TIM7_Init 1 */

  /* USER CODE END TIM7_Init 1 */
  htim7.Instance = TIM7;
  htim7.Init.Prescaler = 0;
  htim7.Init.CounterMode = TIM_COUNTERMODE_UP;
  htim7.Init.Period = 274;
  htim7.Init.AutoReloadPreload = TIM_AUTORELOAD_PRELOAD_ENABLE;
  if (HAL_TIM_Base_Init(&htim7) != HAL_OK)
  {
    Error_Handler();
  }
  sMasterConfig.MasterOutputTrigger = TIM_TRGO_RESET;
  sMasterConfig.MasterSlaveMode = TIM_MASTERSLAVEMODE_DISABLE;
  if (HAL_TIMEx_MasterConfigSynchronization(&htim7, &sMasterConfig) != HAL_OK)
  {
    Error_Handler();
  }
  /* USER CODE BEGIN TIM7_Init 2 */

  /* USER CODE END TIM7_Init 2 */

}
```

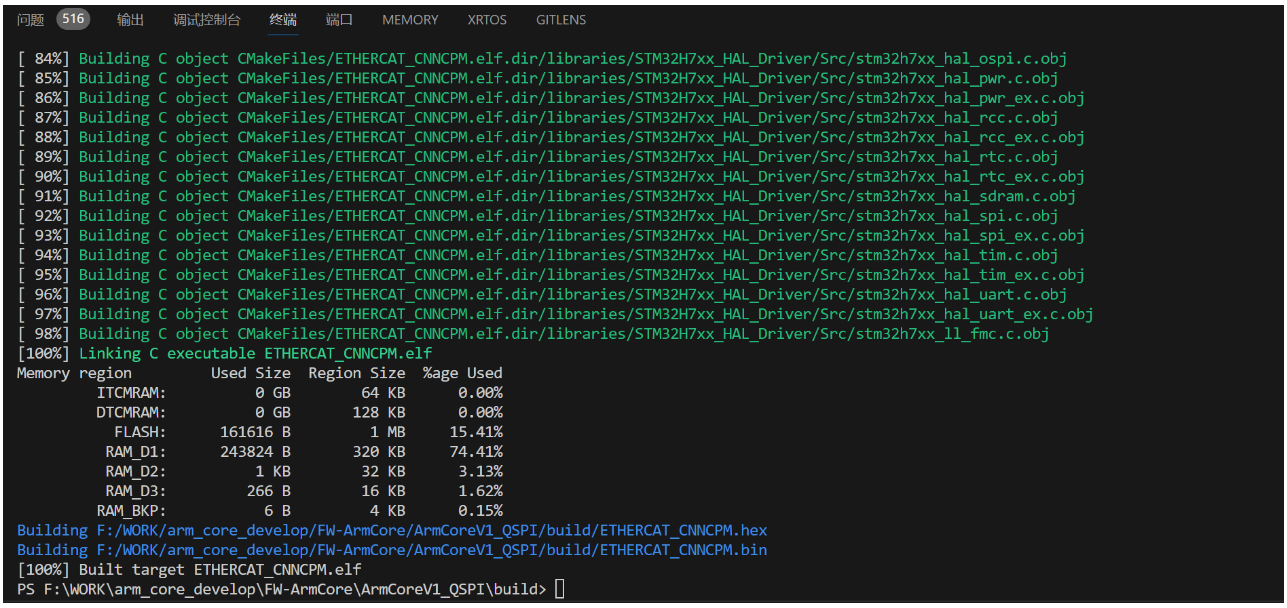
```c
void HAL_TIM_Base_MspInit(TIM_HandleTypeDef* tim_baseHandle)
{

  if(tim_baseHandle->Instance==TIM1) ⋯
  else if(tim_baseHandle->Instance==TIM2) ⋯
  else if(tim_baseHandle->Instance==TIM6) ⋯
  else if(tim_baseHandle->Instance==TIM7)
  {
  /* USER CODE BEGIN TIM7_MspInit 0 */

  /* USER CODE END TIM7_MspInit 0 */
    /* TIM7 clock enable */
    __HAL_RCC_TIM7_CLK_ENABLE();

    /* TIM7 DMA Init */
    /* TIM7_UP Init */
    hdma_tim7_up.Instance = DMA1_Stream6;
    hdma_tim7_up.Init.Request = DMA_REQUEST_TIM7_UP;
    hdma_tim7_up.Init.Direction = DMA_MEMORY_TO_PERIPH;
    hdma_tim7_up.Init.PeriphInc = DMA_PINC_DISABLE;
    hdma_tim7_up.Init.MemInc = DMA_MINC_ENABLE;
    hdma_tim7_up.Init.PeriphDataAlignment = DMA_PDATAALIGN_WORD;
    hdma_tim7_up.Init.MemDataAlignment = DMA_MDATAALIGN_WORD;
    hdma_tim7_up.Init.Mode = DMA_NORMAL;
    hdma_tim7_up.Init.Priority = DMA_PRIORITY_HIGH;
    hdma_tim7_up.Init.FIFOMode = DMA_FIFOMODE_DISABLE;
    if (HAL_DMA_Init(&hdma_tim7_up) != HAL_OK)
    {
      Error_Handler();
    }

    __HAL_LINKDMA(tim_baseHandle,hdma[TIM_DMA_ID_UPDATE],hdma_tim7_up);

  /* USER CODE BEGIN TIM7_MspInit 1 */

  /* USER CODE END TIM7_MspInit 1 */
  }       You, 37分钟前 • Uncommitted changes
}
```

```c
void HAL_TIM_Base_MspDeInit(TIM_HandleTypeDef* tim_baseHandle)
{

  if(tim_baseHandle->Instance==TIM1) ···
  else if(tim_baseHandle->Instance==TIM2) ···
  else if(tim_baseHandle->Instance==TIM6) ···
  else if(tim_baseHandle->Instance==TIM7)
  {
  /* USER CODE BEGIN TIM7_MspDeInit 0 */

  /* USER CODE END TIM7_MspDeInit 0 */
    /* Peripheral clock disable */
    __HAL_RCC_TIM7_CLK_DISABLE();

    /* TIM7 DMA DeInit */
    HAL_DMA_DeInit(tim_baseHandle->hdma[TIM_DMA_ID_UPDATE]);
  /* USER CODE BEGIN TIM7_MspDeInit 1 */

  /* USER CODE END TIM7_MspDeInit 1 */
  }
}

/* USER CODE BEGIN 1 */
/**
  * @brief This function handles DMA1 stream6 global interrupt.
  */
void DMA1_Stream6_IRQHandler(void)
{
  /* USER CODE BEGIN DMA1_Stream6_IRQn 0 */

  /* USER CODE END DMA1_Stream6_IRQn 0 */
  HAL_DMA_IRQHandler(&hdma_tim7_up);
  /* USER CODE BEGIN DMA1_Stream6_IRQn 1 */

  /* USER CODE END DMA1_Stream6_IRQn 1 */
}
```

tim.h

```c
extern TIM_HandleTypeDef htim1;

extern TIM_HandleTypeDef htim2;

extern TIM_HandleTypeDef htim6;

extern TIM_HandleTypeDef htim7;

/* USER CODE BEGIN Private defines */

/* USER CODE END Private defines */

void MX_TIM1_Init(void);
void MX_TIM2_Init(void);
void MX_TIM6_Init(void);
void MX_TIM7_Init(void);

/* USER CODE BEGIN Prototypes */

/* USER CODE END Prototypes */
```

- 重新加载运行cmakelist.txt，以更新makefile文件
- 编译文件，结果如下



```
问题 516  输出  调试控制台  终端  端口  MEMORY  XRTOS  GITLENS

[ 84%] Building C object CMakeFiles/ETHERCAT_CNNCPM.elf.dir/libraries/STM32H7xx_HAL_Driver/Src/stm32h7xx_hal_ospi.c.obj
[ 85%] Building C object CMakeFiles/ETHERCAT_CNNCPM.elf.dir/libraries/STM32H7xx_HAL_Driver/Src/stm32h7xx_hal_pwr.c.obj
[ 86%] Building C object CMakeFiles/ETHERCAT_CNNCPM.elf.dir/libraries/STM32H7xx_HAL_Driver/Src/stm32h7xx_hal_pwr_ex.c.obj
[ 87%] Building C object CMakeFiles/ETHERCAT_CNNCPM.elf.dir/libraries/STM32H7xx_HAL_Driver/Src/stm32h7xx_hal_rcc.c.obj
[ 88%] Building C object CMakeFiles/ETHERCAT_CNNCPM.elf.dir/libraries/STM32H7xx_HAL_Driver/Src/stm32h7xx_hal_rcc_ex.c.obj
[ 89%] Building C object CMakeFiles/ETHERCAT_CNNCPM.elf.dir/libraries/STM32H7xx_HAL_Driver/Src/stm32h7xx_hal_rtc.c.obj
[ 90%] Building C object CMakeFiles/ETHERCAT_CNNCPM.elf.dir/libraries/STM32H7xx_HAL_Driver/Src/stm32h7xx_hal_rtc_ex.c.obj
[ 91%] Building C object CMakeFiles/ETHERCAT_CNNCPM.elf.dir/libraries/STM32H7xx_HAL_Driver/Src/stm32h7xx_hal_sdram.c.obj
[ 92%] Building C object CMakeFiles/ETHERCAT_CNNCPM.elf.dir/libraries/STM32H7xx_HAL_Driver/Src/stm32h7xx_hal_spi.c.obj
[ 93%] Building C object CMakeFiles/ETHERCAT_CNNCPM.elf.dir/libraries/STM32H7xx_HAL_Driver/Src/stm32h7xx_hal_spi_ex.c.obj
[ 94%] Building C object CMakeFiles/ETHERCAT_CNNCPM.elf.dir/libraries/STM32H7xx_HAL_Driver/Src/stm32h7xx_hal_tim.c.obj
[ 95%] Building C object CMakeFiles/ETHERCAT_CNNCPM.elf.dir/libraries/STM32H7xx_HAL_Driver/Src/stm32h7xx_hal_tim_ex.c.obj
[ 96%] Building C object CMakeFiles/ETHERCAT_CNNCPM.elf.dir/libraries/STM32H7xx_HAL_Driver/Src/stm32h7xx_hal_uart.c.obj
[ 97%] Building C object CMakeFiles/ETHERCAT_CNNCPM.elf.dir/libraries/STM32H7xx_HAL_Driver/Src/stm32h7xx_hal_uart_ex.c.obj
[ 98%] Building C object CMakeFiles/ETHERCAT_CNNCPM.elf.dir/libraries/STM32H7xx_HAL_Driver/Src/stm32h7xx_ll_fmc.c.obj
[100%] Linking C executable ETHERCAT_CNNCPM.elf
Memory region         Used Size  Region Size  %age Used
         ITCMRAM:          0 GB        64 KB      0.00%
         DTCMRAM:          0 GB       128 KB      0.00%
           FLASH:      161616 B         1 MB     15.41%
          RAM_D1:      243824 B       320 KB     74.41%
          RAM_D2:          1 KB        32 KB      3.13%
          RAM_D3:         266 B        16 KB      1.62%
         RAM_BKP:           6 B         4 KB      0.15%
Building F:/WORK/arm_core_develop/FW-ArmCore/ArmCoreV1_QSPI/build/ETHERCAT_CNNCPM.hex
Building F:/WORK/arm_core_develop/FW-ArmCore/ArmCoreV1_QSPI/build/ETHERCAT_CNNCPM.bin
[100%] Built target ETHERCAT_CNNCPM.elf
PS F:\WORK\arm_core_develop\FW-ArmCore\ArmCoreV1_QSPI\build>
```

- 接口功能开发：添加TIM7、DMA初始化配置函数，GPIO输出控制接口函数，详见`gpio_imitate.c`和`gpio_imitate.h`文件，并根据需要添加编译文件至cmakelist.txt中